QUANDELA × ETH × Quantum Engineering Commission

Our Challenge FOR YOU – Details

EPFL Hackathon 2026

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ETH Hackathon → MAY 2026

www.quandela.com

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BECOME A PART OF QUANDELA

Become a Quandela Ambassador
& Lead the Next Quantum Generation

QUANDELA KNIGHTS

The Knights

- Certified Quandela community
- Trusted to mentor, supervise & represent
- Local ambassadors

Missions

- Mentor teams
- Run workshops
- Supervise student projects
- Provide feedback
- Help building local community hubs

Incentives

- Paid role for selected events
- Exclusive trainings
- Become part of a global network
- Shape photonic quantum computing

Certified KNIGHT

Join the community & get representative

Complete a short skills assessment

Attend a Quandela challenge or have Perceval & MerLin experience

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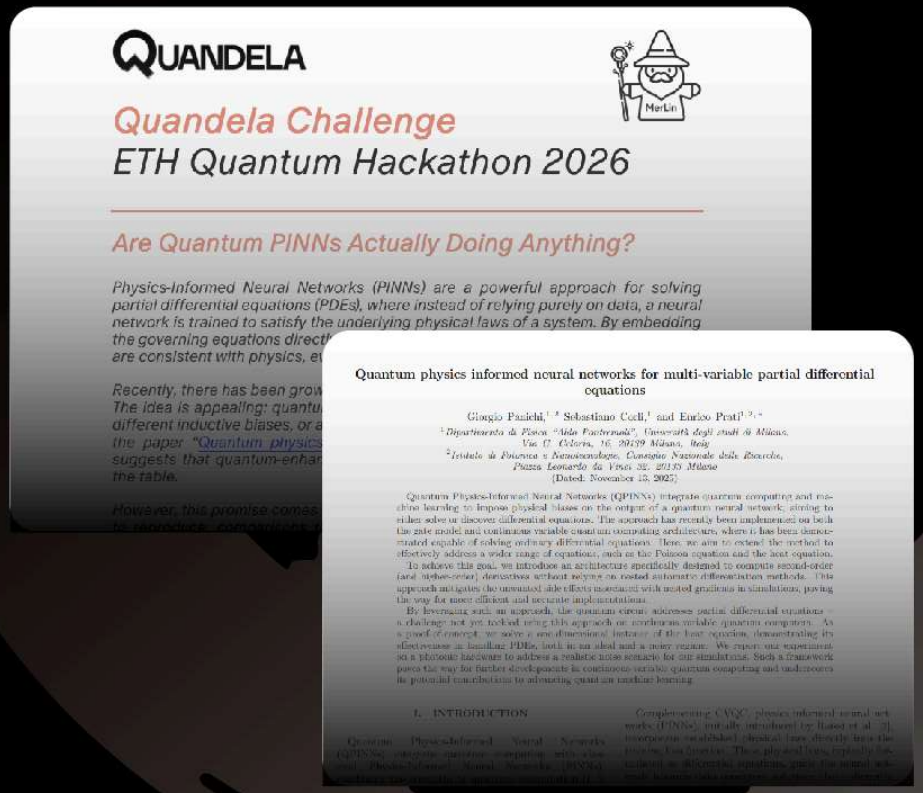
QUANDELA CHALLENGE – ETH HACKATHON 2026

PINN Challenge
Are Quantum PINNs Actually Doing Anything?

You will receive the slides afterwards – all information will be provided as a PDF together with notebooks and incentives!



QUANDELA

Merlin

Quandela Challenge
ETH Quantum Hackathon 2026

Are Quantum PINNs Actually Doing Anything?

Physics-Informed Neural Networks (PINNs) are a powerful approach for solving partial differential equations (PDEs), where instead of relying purely on data, a neural network is trained to satisfy the underlying physical laws of a system. By embedding the governing equations direct
are consistent with physics, e

Recently, there has been grow
The idea is appealing: quantu
different inductive biases, or a
the paper "Quantum physics
suggests that quantum-enha
the table.

However, this promise comes

Quantum physics informed neural networks for multi-variable partial differential equations

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(Dated: November 13, 2025)

Quantum Physics-Informed Neural Networks (QPINNs) integrate quantum computing and machine learning to impose physical biases on the output of a quantum neural network, aiming to either solve or discover differential equations. The approach has recently been implemented on both the gate model and continuous variable quantum computing architecture, where it has been demonstrated capable of solving ordinary differential equations. Here, we aim to extend the method to effectively address a wider range of equations, such as the Poisson equation and the heat equation.

To achieve this goal, we introduce an architecture specifically designed to compute second-order (and higher-order) derivatives without relying on nested automatic differentiation methods. This approach mitigates the unwanted side effects associated with nested gradients in simulations, paving the way for more efficient and accurate implementations.

By leveraging such an approach, the quantum circuit addresses partial differential equations – a challenge not yet tackled using this approach on continuous variable quantum computers. As a proof-of-concept, we solve a one-dimensional instance of the heat equation, demonstrating its effectiveness in familiar PDEs, both in an ideal and a noisy regime. We report our experiment on a photonic hardware to address a realistic noise scenario for our simulations. Such a framework paves the way for further developments in continuous variable quantum computing and underscores its potential contributions to advancing quantum machine learning.

I. INTRODUCTION

Quantum Physics-Informed Neural Networks (QPINNs), emerging quantum computing adaptations of Physics-Informed Neural Networks (PINNs) [1], have recently been implemented on both gate-based and continuous variable (CV) quantum computing architectures [2, 3].

Complementing CVQC, physics-informed neural networks (PINNs), initially introduced by Raissi et al. [4], incorporate established classical laws directly into the training loss function. These physical laws, typically formulated as differential equations, guide the neural network

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PINN Challenge
Background and Motivation

Physics-Informed Neural Networks (PINNs) solve partial differential equations (PDEs) by embedding physical laws directly into the training.

- Work well event with limited or noisy data
- Recent research combines them with quantum models

Hypothesis:
- Richer representations
- Different inductive biases
- Advantage in high-dimensional problems

But:
- Results are hard to reproduce
- Comparisons to classical methods are unclear
- Many quantum models are still simulable

Core question: Is the quantum component actually useful – or just theoretical decoration?

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PINN Challenge
What You Will Do

- Investigate quantum-enhanced PINNs hands-on
- Go beyond theory → **test, reproduce and challenge claims**
- **Focus on:**
  - Performance vs. classical methods
  - Understanding what the quantum part really does
- **Approach:**
  1. Reproduce existing work
  2. Analyze and question it
  3. Improve or redesign

MERLIN

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PINN Challenge
Phase 1: Reproduction

1

Build & Benchmark

- Reimplement the reference QPINN model
- Compare against classical PINN baseline
- Keep fair conditions (similar parameter budgets)
- Evaluate:
  - Accuracy
  - Training behaviour
  - Scaling with circuit depth

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PINN Challenge
Phase 2: Going Beyond

2

Analyze, Redesign, Explore

- Move beyond reproduction → challenge the model design
- Design more meaningful / interpretable circuits
- Test assumptions:
  - Remove auxiliary tricks (e.g. derivative formulations)
  - Evaluate impact on stability & accuracy
- Explore:
  - More expressive circuits (depth, entanglement)
  - Resource cost (not just parameters, but physical cost)

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PINN Challenge
Tools & Resources: What You'll Use

- **MerLin** (Quandela framework)
- **Starter notebooks**:
  - Quantum PINN baseline (MerLin)
  - Classical PINN baseline
- Quandela Cloud (Hub):
  - Run experiments on photonic hardware
  - Study noise & real-world constraints
- **Support**: Quandela Knights on-site during the hackathon and with contact to a Quandela expert for sophisticated questions

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PINN Challenge
What To Submit

Code, quantitative results & visualizations

Presentation summarizing approach, experiments & conclusions

Please upload a folder with the name of your team and which contains all files!

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PINN Challenge
Judging Criteria

Rigor of Investigation

Can the methodology be trusted?

- Controlled, fair experiments?
- Are conlcusions supported by quantitative metrics?
- Are the results consistent and reproducible?

Quantum Insight & Validity

Has the central question been answered?

- Are circuit regimes (depth, expressivity, …) explored?
- Was analyzed when/why the quantum part helps or doesn't?

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PINN Challenge
Judging Criteria

Technical Execution

Was the solution build properly and competently?

- Is the implementation correct, stable and complete?
- Is the classical baseline fair?
- Where MerLin/photonic quantum components used properly?

Clarity & Communication

Are conclusions understandable and believable?

- Were limitations identified?
- Coherent story for hypothesis, experiment and conclusion?
- How well was the presentation?

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Swaptions Challenge
Judging Criteria

Scan for access to the challenger folder with all the resources:

Action step: read the challenge PDF first!

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